

Unified AI-Based Skin Health Decision Support System

Using Image Analysis and Skincare Safety Guidance

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Project Proposal Report

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DECLARATION

We declare that this is our own work, and this proposal does not incorporate without acknowledgement any material previously submitted for a degree or diploma in any other university or Institute of higher learning and to the best of our knowledge and belief it does not contain any material previously published or written by another person except where the acknowledgement is made in the text.

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The supervisor/s should certify the proposal report with the following declaration.

The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of the Supervisor:	2026.03.	
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ABSTRACT

Skin-related conditions such as acne, mild eczema, contact dermatitis, dandruff, diaper rash, warts, cold sores, and minor skin irritations are commonly experienced by individuals across different age groups. Early awareness of these conditions is important for maintaining proper skin care and preventing complications. However, many individuals lack sufficient knowledge to interpret visible skin abnormalities and determine whether a condition requires medical attention.

Recent advancements in artificial intelligence and computer vision have demonstrated strong potential in supporting automated dermatological image analysis. Nevertheless, most existing AI-based dermatology systems primarily focus on detecting severe diseases such as melanoma and skin cancer, while comparatively limited attention has been given to risk-awareness mechanisms for common skin conditions relevant to everyday skincare support.

To address this gap, this research proposes the development of an **AI-based risk-level screening module** that categorizes detected skin lesions into **low-risk, medium-risk, and high-risk groups** based on lesion characteristics identified through image analysis outputs. The proposed system will analyze visual indicators such as abnormal patterns, irregular structures, and severity-related features extracted from lesion identification results.

A key component of this study is the integration of responsible guidance mechanisms that prevent the system from providing direct medical diagnoses. Instead, the module will generate informative alerts encouraging users to monitor their skin condition or seek professional dermatological consultation when higher-risk indicators are detected.

The expected outcome of this research is a reliable risk-awareness mechanism that enhances skin health monitoring while maintaining ethical boundaries between automated decision support and medical diagnosis.

Keywords: Artificial Intelligence, Skin Lesion Risk Screening, Machine Learning, Dermatology Image Analysis, Decision Support System

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LIST OF ABBREVIATION

Abbreviation	Full Form
<i>AI</i>	<i>Artificial Intelligence</i>
<i>ML</i>	<i>Machine Learning</i>
<i>DL</i>	<i>Deep Learning</i>
<i>CNN</i>	<i>Convolutional Neural Network</i>
<i>CV</i>	<i>Computer Vision</i>
<i>HAM10000</i>	<i>Human Against Machine with 10000 Training Images Dataset</i>
<i>ISIC</i>	<i>International Skin Imaging Collaboration</i>
<i>XAI</i>	<i>Explainable Artificial Intelligence</i>
<i>SVM</i>	<i>Support Vector Machine</i>
<i>RF</i>	<i>Random Forest</i>
<i>PCA</i>	<i>Principal Component Analysis</i>
<i>ROI</i>	<i>Region of Interest</i>
<i>GPU</i>	<i>Graphics Processing Unit</i>
<i>API</i>	<i>Application Programming Interface</i>

Abbreviation	Full Form
<i>UI</i>	<i>User Interface</i>
<i>UX</i>	<i>User Experience</i>
<i>IoT</i>	<i>Internet of Things</i>
<i>DB</i>	<i>Database</i>
<i>JSON</i>	<i>JavaScript Object Notation</i>
<i>HTTP</i>	<i>Hypertext Transfer Protocol</i>
<i>HTTPS</i>	<i>Hypertext Transfer Protocol Secure</i>
<i>IDE</i>	<i>Integrated Development Environment</i>
<i>CSV</i>	<i>Comma Separated Values</i>
<i>ROC</i>	<i>Receiver Operating Characteristic</i>
<i>AUC</i>	<i>Area Under the Curve</i>
<i>TP</i>	<i>True Positive</i>
<i>TN</i>	<i>True Negative</i>
<i>FP</i>	<i>False Positive</i>
<i>FN</i>	<i>False Negative</i>

1 INTRODUCTION

1.1 Background and context

Skin conditions are among the most common health issues affecting individuals worldwide. Conditions such as acne, eczema, dermatitis, and other minor skin lesions can significantly impact personal well-being and confidence. Early awareness and appropriate skin care practices are essential to prevent complications and maintain healthy skin.

With the rapid advancement of artificial intelligence and machine learning technologies, automated skin image analysis systems have become an important area of research in healthcare technology. AI-powered systems have demonstrated promising results in analyzing dermatological images and assisting in lesion recognition.

However, most existing AI-based dermatology systems focus on detecting severe diseases such as melanoma or skin cancer. These systems are typically designed for clinical environments and may not be suitable for everyday skin health awareness applications.

Furthermore, when such systems are used outside medical environments, there is a risk that users may interpret automated predictions as medical diagnoses. This creates ethical concerns and highlights the need for safer AI-based solutions that provide supportive guidance rather than diagnostic conclusions.

The proposed research addresses this issue by developing a **risk-level screening mechanism** that converts lesion analysis results into understandable risk categories. The system aims to support early skin health awareness while encouraging professional consultation when necessary.

This research aligns with **United Nations Sustainable Development Goal 3: Good Health and Well-being**, which promotes the use of digital health technologies to improve public health awareness and accessibility.

1.2 Literature Review

Artificial intelligence has become an important research direction in dermatology because skin lesions are highly visual and therefore suitable for computer vision analysis. One of the earliest landmark studies was conducted by Esteva et al. (2017), who demonstrated that deep convolutional neural networks could classify skin lesions at a level comparable to dermatologists. Their research showed that deep learning models can learn complex dermatological patterns directly from image data. However, the primary focus of this study was disease classification rather than providing risk-based awareness for general users [1].

A significant advancement in dermatology research was the introduction of the HAM10000 dataset by Tschandl et al. (2018). This dataset contains a large collection of dermoscopic images of pigmented skin lesions and has become a widely used benchmark for training machine learning models in dermatology. The availability of large-scale datasets has greatly improved the performance of AI-based skin lesion detection systems. However, reliance on benchmark datasets alone may limit the ability of models to generalize to real-world clinical scenarios [2].

Another important dataset contribution is the BCN20000 dataset, which highlights the challenges of applying machine learning models in real-world environments. Research based on this dataset showed that algorithms trained on ideal datasets often experience performance degradation when tested on more diverse and realistic data. This finding emphasizes the need for robust systems capable of handling different imaging conditions and lesion variations, particularly in applications intended for public use [3].

The International Skin Imaging Collaboration (ISIC) challenge further advanced research in automated skin lesion analysis. The ISIC 2018 challenge evaluated algorithms for lesion segmentation, attribute detection, and disease classification. The results showed that although deep learning models achieved promising performance, many systems still produced inconsistent results when applied to new datasets. This highlights the need for reliable evaluation strategies and careful interpretation of AI predictions [4].

Research on human–computer collaboration in dermatology has also shown that artificial intelligence systems can assist medical professionals in improving diagnostic decisions. However, studies have demonstrated that AI systems can also mislead clinicians if their predictions are presented without sufficient explanation or reliability indicators. Therefore, careful design of AI outputs is necessary to ensure safe and effective use in healthcare applications [5].

Another concern identified in recent research is the quality and diversity of dermatology datasets. A systematic review of publicly available skin lesion datasets found that many datasets lack sufficient diversity in terms of skin types, imaging conditions, and patient demographics. This limitation can lead to biased models that perform poorly on underrepresented populations. Addressing these limitations is critical when designing AI systems intended for real-world applications [6].

Explainability has become an important research topic in AI-based dermatology systems. Studies on Explainable Artificial Intelligence (XAI) have shown that deep learning models are often treated as black boxes, making it difficult for users to understand how predictions are generated. Explainable AI techniques aim to improve transparency by identifying important features in images that influence model predictions. These techniques can increase trust and reliability in AI-assisted healthcare systems [7].

Several review studies have analyzed the broader role of artificial intelligence in dermatology. These studies emphasize that AI technologies have the potential to assist dermatologists in diagnosing skin diseases, monitoring patient conditions, and supporting treatment decisions. However, they also highlight the importance of integrating AI systems carefully into clinical workflows to avoid misuse or overreliance on automated predictions [8].

Another important area of research is skin lesion segmentation, which involves identifying the exact boundaries of lesions within an image. Accurate segmentation is essential for extracting meaningful features such as shape, texture, and color patterns. Deep learning methods have significantly improved segmentation accuracy, but the performance of these systems still depends on the quality of training data and annotation consistency [9].

Recent research has explored AI-based decision support systems rather than direct diagnostic models. For example, reinforcement learning approaches have been used to develop systems that assist clinicians in making diagnostic decisions by analyzing patterns and recommending possible actions. These approaches emphasize decision support rather than automated diagnosis, which aligns with safer AI deployment strategies in healthcare [10].

Another important consideration in dermatology AI research is fairness across different skin tones and populations. Studies have shown that many machine learning models perform better on lighter skin types because of limited diversity in training datasets. Addressing these biases is essential to ensure equitable healthcare solutions and improve the reliability of AI-based dermatology tools [11].

Systematic reviews of artificial intelligence for skin cancer detection have highlighted both the strengths and limitations of current technologies. While AI models often demonstrate high accuracy in controlled environments, their safety and reliability in real-world applications remain important concerns. These

studies emphasize the need for responsible system design, clear risk communication, and proper validation before deployment in consumer-facing applications [12].

1.2 Research Gap

Although significant progress has been made in applying artificial intelligence to dermatological image analysis, several challenges remain.

Most existing research focuses on building machine learning models that directly classify skin diseases from dermoscopic images. These systems often attempt to identify specific diseases such as melanoma or benign lesions using deep learning algorithms. While these models achieve high accuracy, their outputs may be misunderstood by general users who are not trained in medical interpretation.

Additionally, many current systems lack a mechanism that translates complex model outputs into simple and understandable information for users. Instead of presenting results in a form that supports decision making, these systems often present disease predictions that may cause confusion or unnecessary anxiety.

Another limitation is the lack of ethical safeguards in many AI-based medical applications. Without clear boundaries between decision support and diagnosis, automated systems may unintentionally encourage self-diagnosis.

Therefore, there is a clear research gap in developing an ethical risk-stratification module that converts lesion detection results into understandable risk levels while maintaining responsible medical boundaries.

The proposed study aims to address this gap by developing a risk-level screening mechanism that categorizes lesions into low, medium, and high-risk groups without providing direct diagnoses.

2 OBJECTIVES

2.1 Main Objective

To develop an AI-based risk-level screening mechanism that categorizes detected skin lesions into low-risk, medium-risk, and high-risk groups in order to support early awareness and encourage responsible dermatological consultation.

3.2 Specific Objectives

- Identify abnormality indicators and severity features in detected skin lesions.
- Define risk stratification criteria for low, medium, and high-risk categories.
- Develop a classification mechanism that maps lesion analysis outputs into risk levels.
- Integrate ethical safeguards that prevent the system from providing diagnostic conclusions.
- Generate responsible alerts advising users to consult a dermatologist when high-risk indicators are detected.
- Evaluate the performance of the proposed model using machine learning evaluation metrics.

3 METHODOLOGY

System Architecture



3. 1 Research Approach

This research follows a Design Science Research (DSR) approach combined with an experimental evaluation methodology. The Design Science approach is appropriate because the objective of this study is to design and develop an intelligent software artifact that addresses a real-world healthcare awareness problem—supporting users in understanding the potential risk level of common skin conditions using artificial intelligence.

The research focuses on developing a risk-level screening module that interprets outputs from skin lesion analysis and categorizes conditions into low-risk, medium-risk, or high-risk groups. The system is intended to provide early awareness and guidance without replacing professional medical diagnosis.

The research process involves the following stages:

- 1) **Problem Identification** - Identifying limitations in existing AI dermatology systems, particularly the tendency to focus on severe disease diagnosis rather than providing safe and understandable risk-awareness guidance for common skin conditions.
- 2) **Design and Development** - Designing a risk-level screening framework that analyzes lesion characteristics and severity indicators extracted from dermatological image analysis.
- 3) **Implementation** - Developing the risk classification mechanism using machine learning techniques and integrating it with the overall skin health decision support system.
- 4) **Evaluation** - Experimentally evaluating the system using established performance metrics such as accuracy, precision, recall, and F1-score to measure the reliability of the risk-level classification.
- 5) **Refinement** - Improving the risk classification model and risk stratification criteria based on evaluation results and experimental findings.

This structured approach ensures that the developed solution effectively addresses the identified research problem while maintaining responsible and ethical use of artificial intelligence

3.2 Technical Methods and Algorithms

The proposed system uses computer vision and machine learning techniques to analyze dermatological images and determine the risk level associated with detected skin conditions.

The system operates as part of a broader AI-driven skin health decision support platform, where image analysis results are interpreted to provide understandable risk-awareness guidance.

Skin Lesion Analysis

The system first processes user-uploaded skin images to identify the most probable skin condition related to common dermatological issues such as acne, mild eczema, contact dermatitis, or minor skin irritations.

The image processing workflow includes the following steps:

1. The input skin image is preprocessed using image processing techniques such as resizing, normalization, and noise reduction.
2. The processed image is analyzed using a **deep learning model based on Convolutional Neural Networks (CNNs)** or transfer learning architectures such as ResNet or EfficientNet.
3. The model extracts visual features from the image and predicts the most probable skin condition category along with associated probability scores.

These prediction outputs are then passed to the **risk-level screening module**, which is the main focus of this research.

Risk-Level Classification Model

The proposed research focuses on designing a risk stratification mechanism that converts lesion identification outputs into interpretable risk categories.

The system evaluates several indicators related to the detected skin condition, including:

- lesion texture patterns
- irregular visual structures
- lesion size and shape characteristics
- prediction confidence levels

These indicators are used to determine the severity level associated with the detected condition.

Machine learning algorithms that may be tested for risk classification include:

- Random Forest
- Support Vector Machine (SVM)
- Neural Network Classifier

The best-performing model will be selected using cross-validation and experimental evaluation.

Risk Stratification Logic

The risk classification mechanism categorizes conditions into three levels:

Low Risk

Indicates minor or common skin conditions with minimal abnormality indicators. Users may be advised to monitor the condition and follow basic skincare practices.

Medium Risk

Indicates conditions that may require monitoring or further observation. Users may be advised to seek professional guidance if the condition persists.

High Risk

Indicates stronger abnormality indicators or higher severity patterns. The system will recommend consulting a dermatologist for further evaluation.

This risk-level approach allows the system to provide responsible guidance while avoiding direct medical diagnosis.

Evaluation Strategy

The effectiveness of the proposed system will be evaluated using standard machine learning performance metrics:

- Accuracy
- Precision
- Recall
- F1-score

Experimental evaluation will be conducted using dermatological image datasets such as **HAM10000 and other relevant skin condition datasets**.

The evaluation results will be used to determine the reliability of the risk-level screening mechanism and improve the system through iterative refinement.

3.3 Tools and Platforms with Justification

The proposed system will be developed using several software tools and machine learning platforms that support image processing, machine learning model development, and system integration.

Tools and Platforms Used for the Development of the Proposed AI-Based Risk-Level Screening System

Tool / Platform	Purpose	Justification
Python	Core programming language	Python is widely used in artificial intelligence, machine learning, and computer vision development due to its rich ecosystem of libraries.

Tool / Platform	Purpose	Justification
TensorFlow / PyTorch	Deep learning framework	Provides efficient tools for building and training deep learning models for skin image analysis.
OpenCV	Image processing library	Used for image preprocessing tasks such as resizing, normalization, and quality assessment.
Scikit-learn	Machine learning algorithms	Used for implementing risk classification models such as Random Forest and Support Vector Machines.
NumPy & Pandas	Data processing and analysis	Used for dataset manipulation and feature analysis during model development.
Jupyter Notebook / Google Colab	Model experimentation environment	Allows rapid experimentation, visualization of results, and GPU-based training for machine learning models.
Flask / FastAPI	Backend API development	Enables integration of machine learning models with the application interface.

These tools are widely used in artificial intelligence research and **provide** reliable, scalable, and efficient development environments for implementing the proposed system.

3.4 Data Collection and Ethical Considerations

The dataset used for this research will be collected from public dermatological image datasets that contain images of common skin conditions.

Possible data sources include:

- HAM10000 skin lesion dataset
- Other publicly available dermatology image datasets
- Research datasets containing common skin condition images

These datasets provide images that can be used to train and evaluate the risk-level screening mechanism.

The collected images will undergo several preprocessing steps, including:

- image resizing
- normalization
- noise reduction
- dataset balancing and labeling

Because dermatological images may contain sensitive visual information, ethical considerations are important during the research process.

The following ethical measures will be implemented:

- No personally identifiable information will be collected from users.
- All datasets used will be publicly available or properly anonymized.
- The system will comply with institutional research ethics guidelines.
- User-uploaded images will not be stored permanently unless required for system evaluation.

Additionally, the system will **avoid providing medical diagnoses** and will instead provide **risk-awareness guidance**, ensuring responsible use of artificial intelligence.

3.5 Validation Strategy and Performance Metrics

The performance of the proposed **risk-level screening system** will be evaluated using experimental validation methods.

The evaluation process will focus on measuring the accuracy and reliability of the **risk classification mechanism**.

Evaluation of Skin Condition Analysis

The image analysis model used to identify skin conditions will be evaluated using common machine learning performance metrics:

- Accuracy – Measures the proportion of correctly predicted conditions.
- Precision – Measures how many predicted conditions are actually correct.
- Recall – Measures how well the system detects relevant skin conditions.
- F1-Score – Provides a balanced measure of precision and recall.

Evaluation of Risk-Level Classification

The risk-level screening module will also be evaluated to measure how effectively it categorizes conditions into **low-risk, medium-risk, and high-risk groups**.

Evaluation metrics include:

- Accuracy of risk classification
- Precision and recall for each risk category
- F1-score for risk prediction performance

These metrics will help determine whether the risk-level screening mechanism provides consistent and reliable results.

Model Validation Strategy

To ensure the reliability and generalization of the developed model, **cross-validation techniques** will be used during training and evaluation.

The dataset will be divided into:

- Training dataset
- Validation dataset
- Testing dataset

This approach helps prevent overfitting and ensures that the model performs well on unseen data.

The results obtained from experimental evaluation will be used to refine the risk-level classification model and improve system performance.

4. HIGH-LEVEL SYSTEM ARCHITECTURE

4.1 Overview of the Proposed Solution and Its Components

The proposed solution is an **AI-driven skin health decision support system** designed to assist users in understanding potential skin condition risks through automated image analysis. The system aims to improve early skin health awareness while ensuring that artificial intelligence is used responsibly without replacing professional medical diagnosis.

At a high level, the system operates as a **multi-component architecture**, where several modules collaborate to analyze uploaded skin images and provide safe, understandable guidance for users. The overall solution consists of the following main components:

1. Image Input Module

This module allows users to upload images of skin conditions through the system interface. The uploaded image becomes the primary input for the analysis process.

2. Image Quality Assessment Module

Before analysis, the system evaluates the quality of the uploaded image. This module checks whether the image meets minimum quality requirements such as brightness, focus, and resolution to ensure reliable analysis results.

3. Skin Lesion Identification Module

This module analyzes the uploaded image using deep learning–based image classification techniques to identify the most probable skin lesion category related to common minor skin conditions such as acne, mild eczema, contact dermatitis, and similar dermatological conditions.

4. Risk-Level Screening Module

This module represents the main focus of this research. It receives the lesion analysis results from the lesion identification component and evaluates the severity and abnormality indicators associated with the detected condition. Based on these indicators, the module categorizes the condition into **low-risk, medium-risk, or high-risk groups**. The goal of this component is to provide interpretable risk awareness while maintaining ethical boundaries that prevent the system from providing medical diagnoses.

5. Confidence-Based Safety Filter

This component evaluates the confidence score produced by the AI model. If the prediction confidence is below a defined threshold, the system prevents unreliable outputs from being presented to the user and instead displays a safety notification requesting a clearer image or professional consultation.

6. Explainability Module (Grad-CAM Visualization)

To improve transparency, the system uses Grad-CAM visualization to generate a heatmap highlighting the image regions that influenced the AI model's analysis. This allows users to understand how the system reached its conclusion.

7. Decision Support / Output Module

The final module presents the system output to the user. The displayed information includes the detected skin condition, the assigned risk level (low, medium, or high), the model confidence score, the explainability heatmap, and advisory messages encouraging appropriate skincare actions or professional consultation when necessary.

4.2 System Diagram of Components

Diagram 1 — Overall High-Level System Architecture

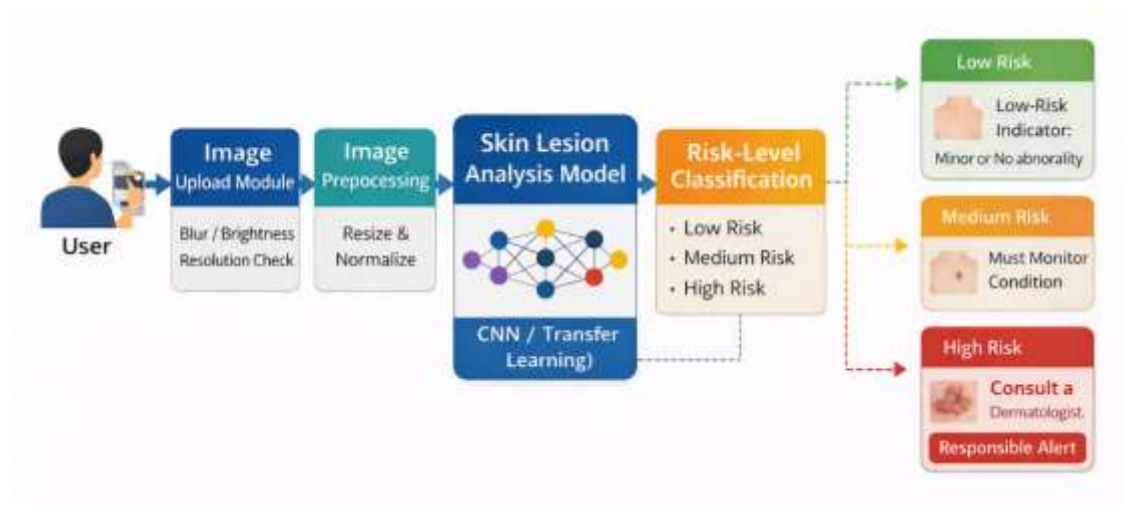


Diagram 2 — Internal Component Structure

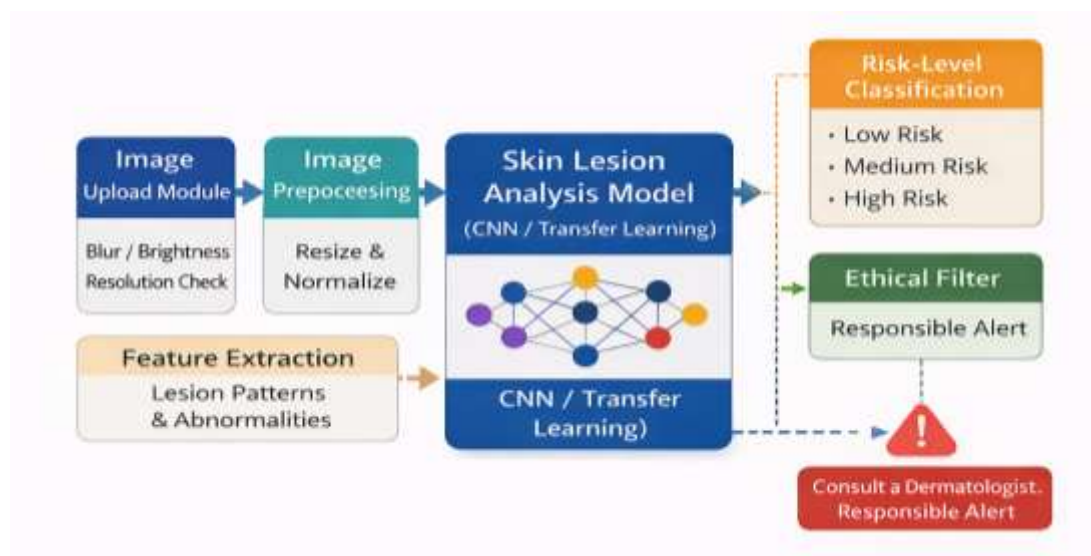
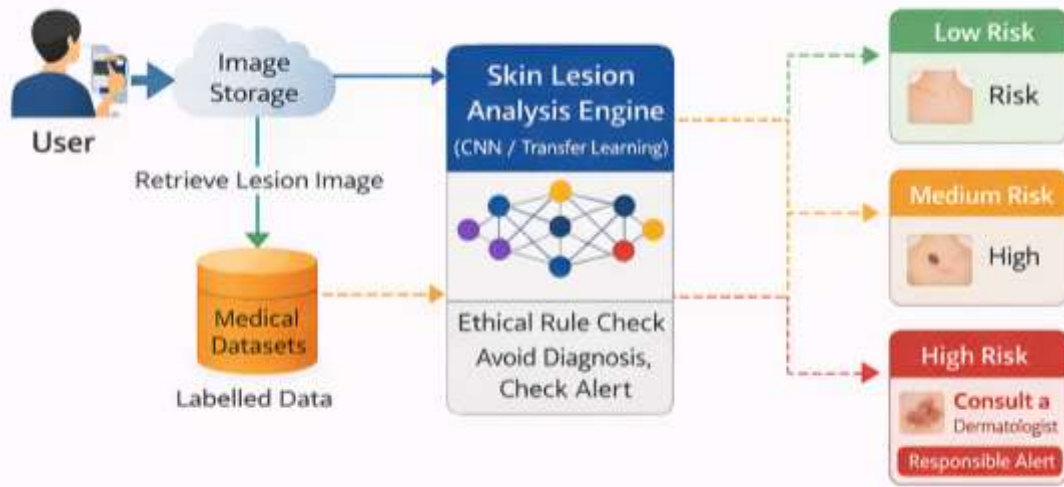


Diagram 3 — Data Flow Diagram



4.3 Individual System Component and Submodules

The individual research component in this study is the **Risk-Level Screening Module**, which is responsible for analyzing lesion identification outputs and categorizing the detected skin condition into **low-risk, medium-risk, or high-risk groups**. This module helps provide interpretable risk awareness to users without generating direct medical diagnoses.

The component contains the following internal parts:

1. Lesion Analysis Input Handler

This unit receives the output from the **Skin Lesion Identification Module**, including the predicted skin condition and associated probability scores. These outputs are used as the primary inputs for the risk evaluation process.

2. Feature and Severity Indicator Extraction Unit

This component analyzes relevant indicators related to the detected skin condition. These indicators may include:

- abnormal visual patterns
- lesion size and structure
- irregular texture patterns
- severity-related characteristics

These indicators help the system determine whether the detected condition appears mild, moderate, or potentially concerning.

3. Risk Stratification Engine

This unit applies predefined **risk stratification rules or machine learning–based logic** to evaluate the extracted indicators. Based on these factors, the system determines the appropriate risk level associated with the detected condition.

4. Risk Classification Unit

After risk evaluation, the system assigns one of the following categories:

- **Low Risk** – Minor or common skin condition with minimal abnormal indicators
- **Medium Risk** – Condition that may require monitoring or further observation
- **High Risk** – Condition showing stronger abnormal indicators that may require professional consultation

5. Confidence Evaluation Unit

This component evaluates the reliability of the risk prediction by analyzing the confidence level associated with the lesion identification output. If the confidence score is below a defined threshold, the system may flag the result as uncertain and advise the user to obtain clearer input or professional advice.

6. Advisory Message Generator

This unit generates responsible guidance messages based on the assigned risk level. Instead of providing medical diagnoses, the system provides advisory information encouraging appropriate skincare actions or professional consultation when necessary.

7. Structured Output Generator

The final stage organizes the results into a structured output format for presentation to the user interface. The output may include:

- detected skin condition
- assigned risk level
- confidence score
- advisory message

4.4 Component Interactions

The components within the risk-level screening module interact in a sequential and controlled manner to ensure reliable and responsible risk assessment.

First, the predicted skin condition and associated probabilities are received from the **Skin Lesion Identification Module**. These outputs are passed to the **Feature and Severity Indicator Extraction Unit**, where relevant lesion characteristics are analyzed.

Next, the extracted indicators are forwarded to the **Risk Stratification Engine**, which applies predefined logic or machine learning techniques to determine the severity level associated with the detected condition.

The resulting evaluation is then processed by the **Risk Classification Unit**, which assigns one of the three predefined risk categories: low risk, medium risk, or high risk.

The classification result is then examined by the **Confidence Evaluation Unit** to verify the reliability of the prediction. If the confidence level is low, the system may generate a caution message advising further observation or professional consultation.

Finally, the **Advisory Message Generator** and **Structured Output Generator** prepare the final system response, which is presented to the user through the system interface.

4.5 Data Flow Explanation

The data flow within the proposed system begins when the **Skin Lesion Identification Module** produces a predicted skin condition and associated probability scores based on the input image.

These outputs are passed to the **Risk-Level Screening Module**, which analyzes the lesion characteristics and severity indicators. The system extracts relevant visual features and evaluates them using predefined risk stratification criteria.

The extracted information is processed by the **Risk Stratification Engine**, which determines the appropriate risk level for the detected condition. This classification is then validated by the **Confidence Evaluation Unit** to ensure the prediction meets reliability requirements.

Following risk classification, the **Advisory Message Generator** produces appropriate guidance messages based on the assigned risk category. The final output includes the detected condition, risk level, confidence score, and recommended user guidance.

This information is then displayed to the user through the system interface and may also be used by other components in the overall skin health decision support system.

4.6 Scalability and Security Considerations

The proposed architecture is designed to support future scalability and system improvements. The modular design of the risk-level screening component allows the system to be expanded in several ways, including:

- incorporating additional skin condition categories
- improving risk evaluation techniques using more advanced machine learning models
- integrating with mobile or web-based healthcare applications
- connecting with additional modules such as skincare recommendations or condition monitoring systems

Because the architecture is modular, updates to individual components can be implemented without redesigning the entire system.

From a security perspective, the system must carefully manage user-uploaded dermatological images, as they may contain sensitive personal information. To ensure data protection, the system should implement:

- secure image upload and storage mechanisms
- restricted access to stored data
- anonymized dataset usage during model training
- minimal retention of user-submitted images where possible

5 USER REQUIREMENTS

5.1 Stakeholder Identification

Stakeholders are individuals or groups who interact with or are affected by the proposed system. Identifying stakeholders helps ensure that the system addresses real user needs while maintaining ethical and safety considerations.

Stakeholder	Description	Role in the System
General Users	Individuals experiencing minor skin concerns seeking skincare guidance	Upload skin images and receive advisory reports
Medical Aestheticians	Professionals providing non-clinical skincare advice	Use the system as a support tool for ingredient guidance
Dermatologists	Medical professionals specializing in skin conditions	Validate system outputs and provide expert consultation when necessary
Researchers / Developers	The research team developing and improving the AI system	Design models, implement modules, and evaluate performance
Cosmetic Product Developers	Organizations developing skincare products	Understand ingredient compatibility and consumer needs
Healthcare Regulators	Regulatory authorities ensuring ethical AI usage	Ensure the system follows safety and non-diagnostic guidelines

Table 5.1 .Stakeholders of the Proposed System

5.2 Functional Requirements

The functional requirements define what the proposed **risk-level screening system** must do.

FR1. The system shall allow users to upload dermatological images for analysis.

FR2. The system shall validate the uploaded image using an image quality assessment process before further processing.

FR3. The system shall reject images that do not meet the required quality standards for brightness, clarity, or resolution.

FR4. The system shall preprocess accepted images by resizing and normalizing them before further analysis.

FR5. The system shall analyze the uploaded image using a lesion identification model to determine the probable skin condition.

FR6. The system shall extract relevant lesion characteristics and severity indicators from the analysis results.

FR7. The system shall apply a risk stratification mechanism to categorize the detected condition into **low-risk, medium-risk, or high-risk groups**.

FR8. The system shall generate advisory messages based on the assigned risk level.

FR9. The system shall display a caution message when the system detects higher-risk indicators or when prediction confidence is low.

FR10. The system shall present the final output to the user including the detected condition, risk level, confidence level, and advisory guidance.

FR11. The system shall log system outputs and evaluation results for research analysis and performance monitoring.

FR12. The system shall support future integration with additional modules in the broader skin health decision support system.

5.3 Non-Functional Requirements

The non-functional requirements define how the system should perform and the quality standards it must satisfy.

Security Requirements

NFR1. The system shall handle uploaded images securely to prevent unauthorized access.

NFR2. The system shall avoid storing personally identifiable information unless explicitly required in future implementations.

NFR3. The system shall ensure secure data storage and controlled access to user-uploaded images.

Performance Requirements

NFR4. The system shall provide risk classification results within an acceptable response time after image upload.

NFR5. The system shall maintain stable performance when processing multiple images during testing or deployment.

NFR6. The system shall achieve acceptable performance metrics such as accuracy, precision, recall, and F1-score for risk-level prediction.

Usability Requirements

NFR7. The system shall provide a simple and user-friendly interface for image upload and result visualization.

NFR8. The system shall present outputs in clear and understandable language suitable for non-technical users.

NFR9. The system shall clearly display safety disclaimers indicating that the output is for **risk awareness and screening support only**, not for medical diagnosis.

Reliability Requirements

NFR10. The system shall avoid generating strong risk predictions when the input image quality is poor.

NFR11. The system shall reduce unreliable outputs by applying confidence evaluation and risk validation mechanisms.

NFR12. The system shall generate responsible advisory messages to guide users toward appropriate actions when higher-risk indicators are detected.

Scalability Requirements

NFR13. The system shall support the future addition of new skin condition categories without requiring major architectural redesign.

NFR14. The system architecture shall support integration with future modules such as skincare recommendations, severity monitoring, or mobile deployment.

Maintainability Requirements

NFR15. The system shall maintain a modular architecture so that components such as preprocessing, risk classification, and advisory generation can be updated independently.

NFR16. The system shall support retraining or improvement of the risk classification model as additional datasets become available.

6. COMMERCIALIZATION PLAN

6.1 Target Market and Customer Persona

The proposed AI-driven **risk-level skin screening system** targets individuals who seek accessible and convenient tools for understanding potential skin condition risks. The primary target market includes young adults, skincare-conscious individuals, and people who frequently experience common minor skin conditions such as acne, mild eczema, contact dermatitis, and other everyday dermatological concerns.

A key customer persona is a young adult or university student aged between **18 and 35** who regularly uses mobile applications for health monitoring, wellness tracking, or skincare guidance. This group often relies on online information or skincare products without professional advice and would benefit from a system that helps them understand whether a skin condition appears to be low risk or requires professional consultation.

Secondary target markets include **skincare clinics, pharmacies, dermatology consultation platforms, and telehealth providers** that may integrate the system into their digital platforms to support preliminary risk screening before professional diagnosis.

6.2 Value Proposition

The main value proposition of the proposed system is to provide an **accessible AI-powered risk-awareness tool** that helps users understand the potential severity of common skin conditions.

Unlike many existing dermatology AI systems that focus primarily on diagnosing severe diseases such as melanoma or skin cancer, the proposed system focuses on **risk-level screening for common skin conditions** that affect everyday skincare decisions.

The system introduces a responsible decision-support approach by categorizing detected conditions into **low-risk, medium-risk, or high-risk levels** rather than providing medical diagnoses. This helps users

interpret potential skin health concerns more safely and encourages appropriate medical consultation when necessary.

By focusing on **risk awareness, user safety, and understandable outputs**, the proposed solution provides an accessible digital skincare support tool suitable for everyday users.

6.3 Revenue Model

If the system is commercialized, several revenue models may be applied. One possible approach is a **freemium mobile application model**, where users can access basic skin risk screening features for free while paying for additional features such as detailed reports, skin condition monitoring history, or advanced skincare guidance.

Another model is a **subscription-based service** that allows users to regularly monitor their skin condition and receive continuous risk-level evaluations.

The system may also be offered through **business-to-business (B2B) partnerships**, where skincare companies, telehealth platforms, or healthcare providers integrate the technology into their digital platforms.

Additionally, the system could be provided through **API-based licensing**, enabling third-party developers or healthcare platforms to integrate the risk screening capability into their own applications.

6.4 Cost Estimation

The development cost of the system mainly involves **software development, machine learning model development, and cloud-based infrastructure**.

Key cost components include:

- Dataset collection and preprocessing for model training
- Cloud computing resources for model training and testing
- Software development tools and frameworks
- Web or mobile application development
- System maintenance and periodic updates

During the research phase, many development tools such as **Python, TensorFlow, and open-source libraries** can be used at minimal cost. However, commercial deployment would require additional infrastructure such as **cloud hosting, secure image storage, and scalable computing resources**.

6.5 Pricing Strategy

The pricing strategy should balance **affordability for users and sustainability for system maintenance**.

For general consumers, a **freemium pricing model** can allow basic risk screening functionality to remain free while offering premium features such as detailed skin analysis reports or long-term monitoring.

For healthcare providers, skincare clinics, or telehealth platforms, **subscription-based licensing or API usage pricing** may be implemented depending on system usage volume or number of users.

This approach ensures that the system remains accessible to everyday users while also supporting commercial sustainability.

6.6 Competitive Advantage

The proposed system offers several advantages compared to many existing AI dermatology solutions.

First, the system focuses specifically on **risk-level awareness for common skin conditions**, rather than attempting to diagnose severe diseases such as skin cancer. This makes the system safer and more appropriate for everyday skincare support.

Second, the system introduces a **risk stratification mechanism** that translates complex machine learning outputs into simple and understandable categories such as low risk, medium risk, and high risk.

Third, the system emphasizes **responsible AI usage** by presenting results as guidance rather than medical diagnoses. This approach reduces ethical concerns and improves user trust.

These features together provide a **more responsible, interpretable, and user-friendly AI skin health support system**.

6.7 Intellectual Property Considerations

If the proposed system demonstrates strong performance and practical usability, it may have potential for intellectual property protection.

The **risk-level screening methodology, system architecture, and decision-support workflow** may form part of a software-based innovation that could be eligible for intellectual property protection depending on jurisdiction.

Additionally, proprietary datasets, trained machine learning models, and software implementations may be protected through **copyright, software licensing, or trade secret protection**.

Before commercialization, it will be important to ensure that all datasets used for training comply with their respective licensing terms and that the system does not use copyrighted data without proper authorization.

7 BUDGET AND BUDGET JUSTIFICATION

The proposed **AI-based risk-level screening system for skin conditions** is primarily a software-oriented research project. Therefore, the required budget is relatively low compared to hardware-intensive projects. Most of the tools used in the development process are open-source technologies, which significantly reduces software licensing costs.

The estimated expenditure mainly includes cloud computing resources for model training, internet connectivity for dataset access and experimentation, and minor costs related to documentation and project preparation.

7.1 Cloud Computing Resources

Item	Description	Estimated Cost
Cloud GPU Resources (Google Colab Pro or similar services)	Used to train machine learning models and perform experiments efficiently using GPU acceleration	LKR 15,000

Table 1: Estimated Cost for Cloud Computing Resources

Justification

Training machine learning models and conducting experiments for the risk-level screening mechanism requires GPU acceleration to reduce training time. Cloud-based GPU services provide cost-effective computational resources without requiring expensive local hardware.

7.2 Internet Connectivity

Item	Description	Estimated Cost
Internet Data	Required for downloading datasets, accessing cloud platforms, and performing online research activities	LKR 5,000

Table 2: Estimated Cost for Internet Connectivity

Justification

Internet connectivity is necessary for accessing dermatological image datasets, running cloud-based machine learning experiments, and conducting literature reviews and research activities.

7.3 Printing and Documentation

Item	Description	Estimated Cost
Printing and Documentation	Printing research reports, proposal documents, and presentation materials	LKR 2,000

Table 3: Estimated Cost for Documentation

Justification

Printing costs are required for preparing physical copies of proposal reports, progress reports, and other documentation needed during project evaluations and presentations.

7.4 Miscellaneous Expenses

Item	Description	Estimated Cost
Miscellaneous Development Expenses	Minor costs related to research activities, presentation preparation, and additional development requirements	LKR 3,000

Table 4: Estimated Miscellaneous Expenses

Justification

Miscellaneous expenses may arise during the research process, including preparation of presentations, minor software utilities, or other small research-related requirements.

7.5 Total Estimated Budget

Category	Estimated Cost
Cloud Computing Resources	LKR 15,000
Internet Connectivity	LKR 5,000
Printing and Documentation	LKR 2,000
Miscellaneous Expenses	LKR 3,000

Estimated Total Budget: LKR 25,000

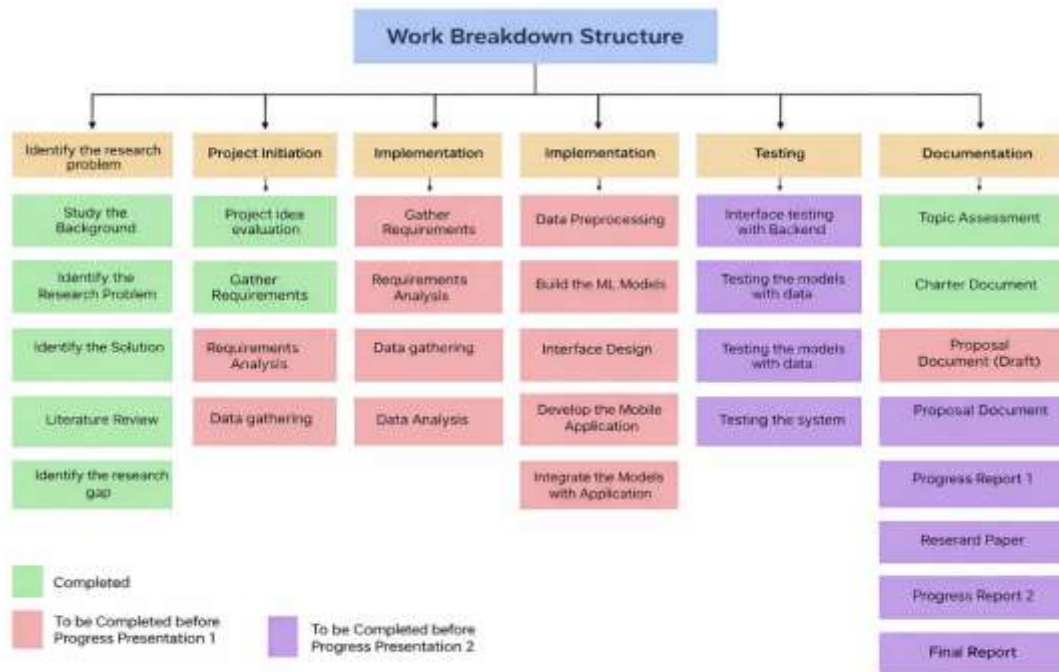
8 WORK BREAKDOWN STRUCTURE

The Work Breakdown Structure (WBS) defines how the project tasks are distributed among the research team members. The purpose of the WBS is to ensure balanced workload distribution, clear responsibility allocation, and proper timeline alignment for successful completion of the research project. The project activities are divided into several key phases including research, system design, development, integration, testing, and documentation.

8.1 WBS among Team Members

Member	Student ID	Responsibility Area
Member 1	IT22100634	Skin Image Analysis & Condition Detection System
Member 2	IT22179494	Skin Risk Level Identification & Prediction System
Member 3	IT22099068	AI-Based Ingredient Recommendation System with Skincare Safety & Decision Support

8.2 Task Breakdown per Member



The work breakdown structure for this project consists of five main stages: identification of the research problem, project initiation, implementation, testing, and documentation. In the first stage, the research problem related to automated skin lesion risk screening will be studied and analyzed in order to clearly understand the objectives and requirements of the proposed system. This stage also includes studying the background, conducting a literature review, and identifying the research gap in existing AI-based dermatology systems.

The project initiation stage focuses on evaluating the project idea and gathering system requirements. During this stage, relevant datasets of skin lesion images are collected and analyzed to support the development of the machine learning model.

The implementation stage involves preprocessing the dataset, developing machine learning models for lesion analysis, and building the risk classification module that categorizes detected skin lesions into low-risk, medium-risk, and high-risk groups. The developed model will then be integrated into the overall skin health decision support system.

The testing stage focuses on evaluating the performance of the developed model using testing datasets and machine learning evaluation metrics. System testing will also be conducted to ensure that the risk-level screening mechanism functions accurately and reliably.

Finally, the documentation stage includes preparing the proposal report, progress reports, research paper, and the final project report, along with all necessary technical documentation related to the system development.

8.3 Project Timeline

Phase	Activities	Timeline
Phase 1	Literature Review & Problem Analysis	Weeks 1 – 3
Phase 2	Dataset Identification, Collection & Preprocessing	Weeks 4 – 6
Phase 3	System Architecture Design	Weeks 7 – 8
Phase 4	Skin Lesion Analysis Model Integration	Weeks 9 – 11
Phase 5	Risk-Level Screening Module Development	Weeks 10 – 12
Phase 6	Risk Classification Logic & Advisory Message Generation	Weeks 11 – 14
Phase 7	System Integration	Weeks 15 – 16
Phase 8	System Testing and Validation	Weeks 17 – 18
Phase 9	Final Documentation & Presentation	Weeks 19 – 20

8.4 Balanced Workload Distribution

The workload is distributed among the team members to ensure that each member contributes equally to the development of the proposed system.

Member 1 – Skin Image Analysis & Condition Detection System

Responsibilities include:

Dermatological image dataset preparation

Image preprocessing implementation

Skin condition classification model development

AI model training and evaluation

Model performance analysis

Member 2 – Skin Risk Level Identification & Prediction System

Responsibilities include:

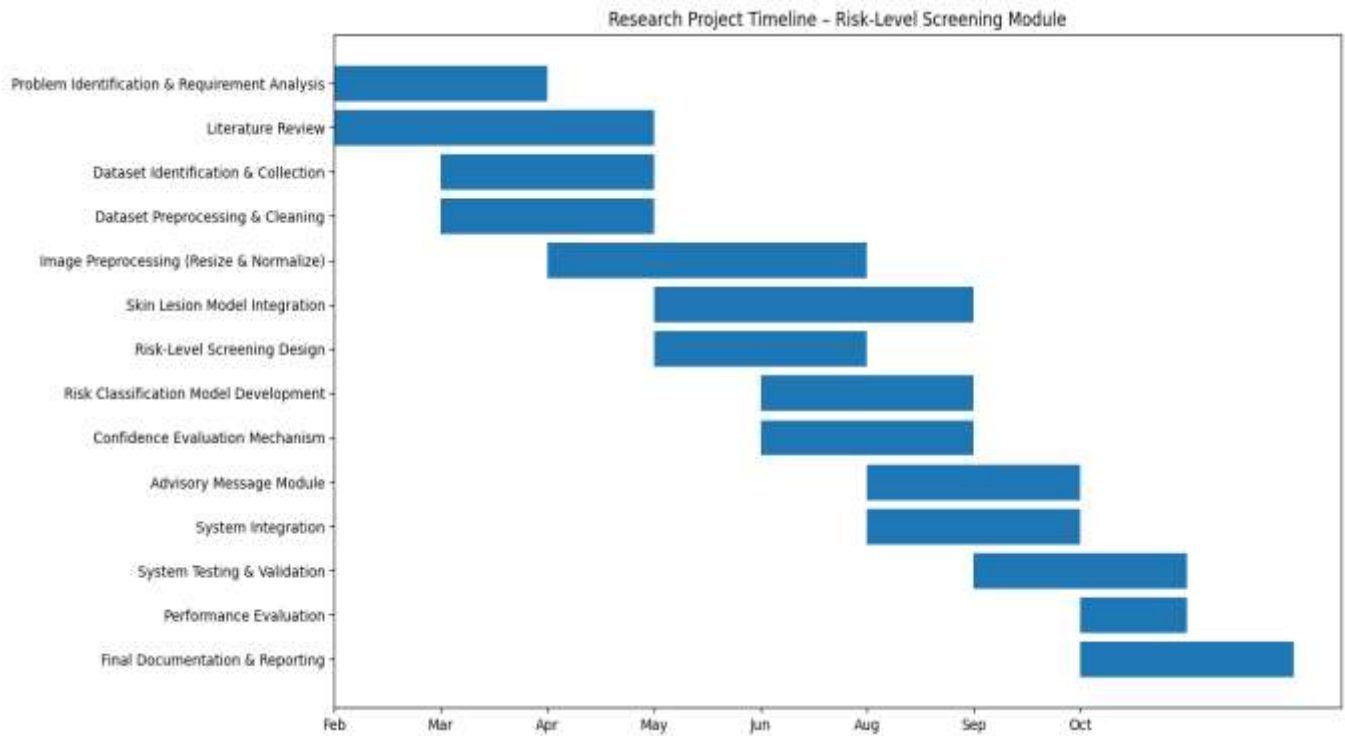
Skin risk assessment research
Risk prediction algorithm development
Risk level classification (Low, Medium, High)
Severity prediction model development
Risk analysis validation and testing

Member 3 – AI-Based Ingredient Recommendation System with Skincare Safety & Decision Support (*Your Component*)

Responsibilities include:

Ingredient knowledge base creation
Skin condition to ingredient mapping logic
Ingredient recommendation engine development
Ingredient safety rule implementation
Ingredient compatibility and conflict detection
Skincare decision support logic development
Recommendation explanation generation
System integration and validation

9 GANTT CHART



10 REFERENCES

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11 APPENDIX

Appendix A – Survey Instrument

A preliminary survey will be conducted to understand user awareness of skin health, common skincare habits, and expectations from an **AI-based skin risk awareness system**. The collected responses will help guide the design of the risk-level screening mechanism and ensure that the system provides useful and understandable guidance for users.

Sample Questions

1. What is your age group?
2. How often do you experience minor skin problems such as acne, dryness, irritation, or mild eczema?
3. Do you usually consult a dermatologist when you notice changes in your skin condition?
4. Have you ever used a mobile application or online platform for skincare guidance?
5. Would you use an AI-based system that analyzes skin images and provides risk-level awareness?
6. What type of features would you expect from such a system (image analysis, risk-level warnings, skincare guidance, safety alerts)?

Appendix B – Detailed System Diagrams

This appendix includes technical diagrams used to explain the proposed system architecture and workflow.

The diagrams may include:

- High-Level System Architecture
- Internal Component Structure of the Risk-Level Screening Module
- Data Flow Diagram (DFD)
- AI Processing Pipeline

These diagrams illustrate how the system performs **skin image analysis, risk-level classification, and decision-support guidance generation**.

Appendix C – Preliminary UI Mockups

The system will provide a **user-friendly interface** designed to help users easily analyze skin conditions and understand the risk level of detected abnormalities.

The interface may include the following screens:

1. Login / Registration Screen
2. Skin Image Upload Screen
3. Image Analysis and Risk-Level Result Screen
4. Risk Awareness Guidance Dashboard
5. Final Skin Health Advisory Report Screen

These screens are designed to ensure that users can easily upload images and clearly understand the system output.

Appendix D – Risk Analysis

Risk	Description	Mitigation
Data Quality Issues	Poor image quality may affect system analysis results	Implement image preprocessing and quality validation mechanisms
Model Errors	Machine learning models may incorrectly classify risk levels	Use larger datasets and performance evaluation techniques
User Misinterpretation	Users may assume the system provides medical diagnosis	Provide clear disclaimers stating that the system is for risk awareness only
Privacy Concerns	Handling user-uploaded skin images may involve sensitive data	Use secure storage, encryption, and minimal data retention

Appendix E – Ethical Considerations

The proposed system follows several important ethical principles to ensure responsible use of artificial intelligence.

User Privacy Protection

User-uploaded images will be handled securely to prevent unauthorized access or misuse.

Non-Diagnostic Disclaimer

The system provides **risk-awareness guidance only** and does not replace professional medical diagnosis.

Data Confidentiality

User data and images will not be shared with third parties without proper authorization.

Responsible AI Usage

Efforts will be made to reduce bias in the machine learning model and ensure fair, transparent, and responsible AI predictions.